PSYCON: CONTEXTUAL COMMUNICATION BEHAVIOR AND EVIDENCE-LINKED PSYCHOLOGIST OBSERVATIONS

SEPARATE TWO-PAGE RESEARCH SYNOPSIS

Arjun Vijay Prakash **Saksham Yadav**

October 12, 2026

ABSTRACT

Group-discussion behavior depends on what happened immediately before, whether a participant had a fair opportunity to respond, and the person's earlier behavior. A brief contribution or an overlapping turn alone cannot establish anxiety, disengagement, or personality. PSYCON investigates whether speaker-specific conversational measurements can predict contextual observations reviewed by psychologists, and whether earlier personal history improves evidence-supported communication reports. Its local research instrument processes audio, produces anonymous speaker intervals and transcripts, retains timestamped exchanges, and connects reviewed participant mappings to a versioned A–T marksheet. Ratings distinguish 0–4 observations, insufficient opportunity, and missing answers. The planned behavioral evaluation measures independent rater agreement, held-out per-item prediction error, and blinded report grounding and usefulness. Participant/source grouping prevents related recordings from crossing evaluation boundaries. The implemented report comparison uses transcript-only, structured-context, and full-PSYCON inputs with matched local model settings. Human behavioral outcomes remain pending. Retained engineering evidence covers eleven recordings and 85 person-recording slots: guarded PSYCON yielded 57 ready profiles and guarded Nemotron yielded 51 in each tested configuration. These describe coverage, not identity accuracy or psychological validity. The 76 imported version 5.0 forms contain 750 numeric, 1,662 N/O, and twenty missing answers. A separate exploratory Docker fit trained fourteen source-score targets using 423 real labels; independent validation is unavailable. The contribution is an auditable behavioral measurement workflow. The next study must establish permission, reproducible human scoring, and independent evaluation before claims of predictive or communication benefit.

1 Introduction and behavioral question

The question is whether measurements and context predict reviewed A–T observations on held-out people/sources, and whether earlier history improves supported reports. The rubric concerns discussion tracking, contribution structure, turn-sharing, challenge, and delivery change. These are operational areas, not diagnostic factors. Reduced speech after challenge needs the event, earlier behavior, later response, and opportunity. Source context and measurement validity matter because a plausible model description cannot supply its own human validation.

2 Innovation and objectives

The contribution links participant, source time, exchange, reviewer, and revision in an auditable measurement workflow. H1 predicts lower held-out item error than the training-median baseline. H2 predicts better blinded grounding/usefulness without increased overclaiming. A history-only experiment must isolate history from the evidence/references also added by full PSYCON.

3 Methodology and controls

The unit is a participant within a discussion. A reviewer confirms person/participant/speaker mapping. Nineteen author-supplied cross-recording links reduce 85 slots to at most 66 people and impose six provisional recording groups; further overlaps remain unknown. Linked recordings must share a partition. Recordings retain context, time, permission, and hashes. Audio processing conservatively attributes speech and preserves evidence; timing/marker proxies do not prove intent.

Psychologists supply scores, opportunity, confidence, recording support, and evidence, followed by independent review. Disagreement remains available before adjudication. Positive scores need timestamps; event-linked items need event/response windows, and Q–T need an earlier same-session baseline. Item S lacks predictors; visual-only T cannot train the audio model. Version 5.0 imports remain separate from v4.0; transcription and validity review are pending.

Frozen datasets group repeated people and related sources before splitting. Training-only preprocessing precedes per-item logistic classification; validation selects regularization, with test data separate. Fitting minimums are engineering gates, not a power calculation. Recruitment, sample size, and prior approval remain to be documented. Analysis needs item error against the training-median baseline, weighted agreement, exclusions, abstentions, and grouped uncertainty. Reports need blinded ratings, randomized order, and a primary outcome. Language, topic, moderation, opportunity, and recording quality are confounds.

4 Current results, limits, and next study

The earlier audiovisual comparison retained 57 PSYCON profiles versus 51 per Nemotron mode out of 85 slots, not unique people. An exploratory source-score fit used forty examples, nine recordings, five known groups and 423 labels across G, I and P1–P12. All examples trained the original models; independent test accuracy is unavailable. Training agreement was 97.9%; omitted-group agreement was 57.2% on 402 scored labels versus median-reference 64.4%, with MAE 0.562 versus 0.460. Earlier retrospective repairs gave 60.9% and 56.5% agreement. A fixed twelve-candidate compact/ordinal procedure then reached 62.2%, MAE 0.490 and class sensitivity 30.0% on the same 402 labels, still below median-reference agreement/error. Score-2 recall remained zero and score 0 was unobserved. Fourteen final fits, eight population references and six feature models, remain exploratory candidates. No repaired procedure was promoted. The pilot does not settle H1. Reporting now separately provides deterministic measurements and an explicitly untested goal exercise; bounded local generation reproduces supplied quantities and withholds invalid or duplicate claims. No eligible history or blinded usefulness result establishes H2. Strict behavioral eligibility remains zero because opportunity, confidence, evidence and independent review are missing. N/O and missing answers were not invented.

Next work must verify transcriptions, permission/mapping, and independent rater reliability, then freeze an independent sample. A history-only ablation should hold transcript, evidence, references, model, and decoding fixed. Poor agreement, no baseline improvement, or increased unsupported claims would challenge the hypotheses. Hardware/3D/KiCad work is paused future measurement research. Voluntary communication review is a potential use; diagnosis, disciplinary scoring, causation, and worldwide benefit are unsupported.

5 Acknowledgements and selected references

School psychologists supplied observation reference material; open-source authors supplied speech tools. Private identities and school/location identifiers are omitted. AI assisted this working draft; authors must independently rewrite applicable submission text and verify claims, references, contributions, consent, and approvals. No poster is included.

Sacks, Schegloff and Jefferson (1974), turn-taking, doi:10.2307/412243. Cronbach and Meehl (1955), construct validity, doi:10.1037/h0040957. Cohen (1968), weighted agreement, doi:10.1037/h0026256. Full references and engineering sources accompany the separate paper. Study framing follows the ISEF science judging criteria; human-study requirements follow the Human Participants rules.